

Data Collection & Extraction from Database

Objective

The objective of this phase is to collect reliable agricultural crop production data and prepare it for analysis in Tableau. The collected dataset serves as the foundation for generating meaningful insights into crop production, seasonal trends, and regional agricultural performance across India.

Description

Data collection is the process of gathering, verifying, and organizing agricultural information from trusted sources. The dataset contains historical crop production records from **1997 to 2021**, including information such as state, district, crop name, season, cultivated area, and production.

After collecting the data, it was cleaned and transformed to remove inconsistencies, missing values, and duplicate records. The prepared dataset was then connected to Tableau for further visualization and dashboard creation.

Data Source

- Historical Agricultural Crop Production Dataset (1997–2021)
- Dataset Format: CSV/Excel
- Source: Public agricultural data repository

Data Attributes

The dataset includes the following important fields:

- State Name
- District Name
- Crop Name
- Crop Year
- Season
- Area Cultivated (Hectares)
- Production (Tonnes)
- Yield

Data Collection Process

1. Download the agricultural dataset.
2. Verify the quality and completeness of the data.
3. Remove duplicate records.
4. Handle missing or inconsistent values.

5. Standardize column names and data formats.
6. Save the cleaned dataset in CSV/Excel format.
7. Import the dataset into Tableau.

Database Connection in Tableau

The cleaned dataset was connected to Tableau using the following steps:

1. Open Tableau Desktop.
2. Select **Microsoft Excel** or **Text File (CSV)** as the data source.
3. Load the agricultural dataset.
4. Verify field names and data types.
5. Connect the dataset successfully.
6. Begin creating worksheets and dashboards.

Expected Outcome

After completing this phase:

- A clean and structured agricultural dataset is available.
- The data is successfully connected to Tableau.
- The dataset is ready for visualization and analysis.
- Reliable insights can be generated through interactive dashboards.

The screenshot displays the Skill Wallet dashboard for a project titled "India's Agricultural Crop Production Analysis". The dashboard is divided into several sections:

- Left Sidebar:** Contains navigation links for "Dashboard", "Skill Bank", "Skill Wallet" (highlighted), "Skill Card", and "Skill Credits".
- Header:** Shows the project title "India's Agricultural Crop Production Analysis" and tabs for "Team", "Overview", "Workspace" (selected), and "Kanban".
- Main Content Area:**
 - Data Collection & Extraction from Database:** A section with a purple "Epic" tag, "Complexity: Medium", and "Duration: 5h 0m". It includes a "Description" box stating: "Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, evaluate outcomes and generate insights from the data." Below this is a "Resources" section with the text "No resources available".
 - Project Progress:** A section showing a progress bar at 90%. It includes buttons for "View Demo" and "View Repo". Below the progress bar is a list of tasks with their completion status:

Task	Progress
Data Collection & Extraction from Database	90%
Downloading the Dataset	
Connect Dataset with Tableau	
Data Preparation	90%
Data Visualization	90%
Dashboard	90%